INCH-POUND

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MILITARY SPECIFICATION

PUMP UNIT, CENTRIFUGAL: PORTABLE, SELF-PRIMING, ENGINE-DRIVEN (DEWATERING; CONTRACTOR AND TRASH TYPE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of portable, self-priming, power-driven, radial flow, single suction, single stage, centrifugal pump units.

* 1.2 Classification. The pump units are of the following types and sizes, materials, classes, and styles, as specified (see 6.2):

TYPE

Type I - N, medium head, dewatering contractor pump.

Size 5M - 1.5 inch; 5,000 gallons per hour (gph).
Size 8M - 2.0 inch; 8,000 gph.
Size 10M - 2.0 inch; 10,000 gph.
Size 15M - 3.0 inch; 15,000 gph.
Size 18M - 3.0 inch; 18,000 gph.
Size 20M - 3.0 inch; 20,000 gph.
Size 40M - 4.0 inch; 40,000 gph.
Size 90M - 6.0 inch; 90,000 gph.
Size 125M - 8.0 inch; 125,000 gph.
Size ZOOM - 10.0 inch; 200,000 gph.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4320

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Type III - MT, solids handling, trash pump

Size 5MT - 1.5 inch; 5,000 gph.
Size 10MT - 2.0 inch; 10,000 gph.
Size 18MT - 3.0 inch; 18,000 gph.
Size 33MT - 4.0 inch; 33,000 gph
Size 35MT - 4.0 inch, 35,000 gph.
Size 70MT - 6 0 inch, 70,000 gph

Type IV - MTC, compact, trash pump.

Size 5MTC - 1.5 inch; 5,000 gph.
size 9MTC - 2.0 inch; 9,000 gph.
Size 15MTC - 3.0 inch; 15,000 gph.
Size 22MTC - 4.0 inch; 22,000 gph.

MATERIAL

Material A - Cast iron casing; ferrous impeller.
Material B - Cast iron casing; bronze impeller.
Material C - Aluminum casing; ferrous impeller.
Material D - Aluminum casing; bronze impeller.

CLASS

Class 1 - Base-mounted with carrying handle or lifting eye.
Class 2 - Wheel-mounted, two-wheel, manually-propelled.
Class 2A - Wheel-mounted, two-wheel, vehicuiar-towed.
Class 3 - Skid-mounted.
Class 4 - Wheel-mounted, four-wheel, wagon-type, vehicular-towed.
Class 4A - Wheel-mounted, four-wheel, tandem-axle with suspension system, vehicular-towed.
class 5 - Base-mounted with tubular, wrap-around, roll-bar frame.

Note : See table I for applicability of class to size.

STYLE

Style A - Gasoline-engine-driven air- or liquid-cooled.
Style B - Diesel-engine-driven liquid-cooled.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2 1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

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SPECIFICATIONS

FEDERAL

W-F-800 - Fuel Oil, Diesel.

MILITARY

MIL-P-10603 - Pumps and Pumping Units, Centrifugal, Power-Driven, for Water; Packaging of.

STANDARDS

FEDERAL

FED-STD-H28/10 - Screw Threads Standards for Federal Services, Section 10, American National Hose Coupling and Fire-Hose Coupling Threads.
FED-STD-297 - Rustproofing of Commercial (Nontactial_ Vehicles.
FED-STD-595 - Colors Used in Government Procurement.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-1400 - Engines, Gasolime and Diesel, Methods of Test.
MIL-STD-1414 - Noise Limits for Army Material.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Title 29, CFR, 1910.95 - Occupational Noise Exposure.
Title 29, CFR, 1910.219 - Mechanical Power-Transmission Apparatus.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations (FMVSSR).

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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ENVIRONMENTAL PROTECTION AGENCY (EPA)

Title 40, CFR, Part 86 - Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines: Certification and Test Procedures.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402)

2.2 Non-Government publication. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which are current on the date of the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B 1.20.1 - Pipe Threads, General Purpose (Inch).
ASME B 16.1 - Cast Iron Pipe Flanges and Flanged Fittings.
ASME B 16.5 - Pipe Flanges and Flanged Fittings.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

ASTM

ASTM A 48 - Gray Iron Castings.
ASTM A 126 - Gray Iron-Castings for Valves, Flanges, and Pipe Fittings.
ASTM A 536 - Ductile Iron Castings.
ASTM B 26/B26M - Aluminum-Alloy Sand Castings.
ASTM B 62 - Composition Bronze or Ounce Metal Castings.
ASTM B 108 - Aluminum-Alloy Permanent Mold Castings.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

CONTRACTORS PUMP BUREAU (CPB)

Contractors Pump Manual.

(Application for copies should be addressed to the Contractors Pump Bureau, 13975 Connecticut Avenue, Suite 310, Silver Springs, MD 20906.)

HYDRAULIC INSTITUTE (HI)

Hydraulic Institute Standards.

(Application for copies should be addressed to the Hydraulic Institute, 712 Lakewood Center North, 14600 Detroit Ave, Cleveland, OH 44107.)

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SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J534 - Lubrication Fittings.
SAE J1349 - Engine Power Test Code - Spark Ignition and Compression Ignition - Net Power Rating.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Suite 304, Akron, OH 44313.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2 3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. Each pump unit shall consist of an integrally self-priming centrifugal pump as specified in the Hudraulic Institute Standards, a direct-connected internal combustion engine, and a common base, skid, or wheel mounting. The pump capacity in gph listed in 1.2 to identify pump sizes shall be interpreted as the nominal capacity. The size of the pump listed in 1.2 indicate suction and discharge openings, which shall be of identical dimension on each size, The applicability of class to size shall be as specified in table I, as represented by an “X” in the column under the size of pump.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.4),

3.3 Codes and standards. The pump shall conform to the requirements of the CPB Contractors Pump Manual as applicable,

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or -maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term “recovered materials” means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

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TABLE I. Applicablity of class to size.

	Size - Type I									
Class	5M	8M	10M	15M	18M	20M	40M	90M	125M	200M
1	X	X	X	X	X	-	-	-	-	-
2	X	X	X	X	X	X	X	-	-	-
2A	-	-	-	-	-	-	X	X	-	-
3	-	-	-	-	-	X	X	X	X	X
4	-	-	-	-	-	-	-	X	X	X
4A	-	-	-	-	-	-	-	X	X	X
5	X	X	-	-	-	-	-	-	-	-

	Size - Type III					
Class	5MT	10MT	18MT	33MT	35MT	70MT
1	X	X	X	-	-	-
2	X	X	X	-	-	-
2A	-	-	-	X	X	X
3	-	-	-	X	X	X
4	-	-	-	-	X	-

	Size - Type IV			
Class	5MTC	9MTC	15MTC	22MTC
1	X	X	X	-
2	X	X	X	X
3	-	-	-	X
5	X	X	-	-

3.4.1 Iron castings. Gray cast iron used for pump casing impellers, and wera plates shall meet the requirements of ASTM A 48, class No. 25 or better, or of ASTM A 126, class B or better. Ductile (nodular) iron of ASTM A 536, or malleable from castings may be used in lieu of gray iron castings provided the surface has a Brinell hardness number of not less than 160.

3.4.2 Steel. Steel used for such components as pump shafts, impellers, wear plates, skids, bases, and diffusers shall be of a type and grade normally used by the contractor on his standard commercial units.

3.4.3 Aluminum. Aluminum for pump casing shall be a silicon-copper-aluminum alloy or a silicon-magnesium-aluminum alloy conforming to ASTM B 26/B26M for sand castings or ASTM B 108 for permanent mold castings, as applicable. Pressure-containing castings shall be heat-treated as required to produce a tensile strength of not less than 30,000 pound-force per square inch (psi) . Acceptable alloys shall include, but not necessarily be limited to, commercial alloys 319.0 and 356.0.

3.4.4 Bronze. Bronze castings for pump impellers, when material B or D is specified, shall be composition bronze (alloy 85-5-5) conforming to ASTM B 62.

3.5 Desire and construction. The pump unit shall be constructed to prevent conditions hazardous to personnel or deleterious to equipment and shall permit easy accessibility for maintenance and service in the field. The pump unit shall withstand the strains, jars, vibrations, and other conditions normal to shipping, storage, and installation; and shall be complete so that, when installed, it can be used for any operation for which it was designed.

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3.5.1 Pump. The pump shall be of the self-priming, radial flow, single suction, single stage, centrifugal-type referred to commercially as contractor pumps . Each pump shall consist of a casing, impeller, seal, connections, and a diffuser ring or air separation chamber. The impeller shaft may be an extension of the engine crankshaft except for 4.0 inch and larger pumps, which shall be furnished with a separate bearing-mounted impeller shaft, connected to the engine crankshaft by a flexible-coupling.

3.5.1.1 Type I pump . Type I pump shall be designed and constructed to meet the requirements of the CPB Contractors Pump Manual for M rated pumps, as well as all applicable requirements specified herein. Type I pump shall be equipped with an open or semi-open impeller designed to pass solids equal to 10 percent by volume and equal to 25 percent of the nominal inlet diameter of the pump.

3.5.1.2 Type III pump. Type III pump shall be designed and constructed to meet the requirements of the CPB Contractors Pump Manual for MT rated pumps, as well as all applicable requirements specified herein. Type III pumps shall be similar to type I pumps except they shall be designed to incorporate a removable end plate which shall provide ready and easy access to the interior as well as to the impeller for clean-out purposes Type III pumps shall be designed to pass spherical solids in the sizes shown in table II.

TABLE II. Solids passing sizes for type III pumps.

Pump rating	Pump size (in inches)	Solids size (in inches)
5MT	1.5	1.00
10MT	2.0	1.25
18MT	3.0	1.50
33MT	4.0	2.00
35MT	4.0	2.00
70MT	6.0	2.50

3.5.1.3 Type IV pump . Type IV pump shall be designed and constructed to meet the requirements of the CPB Contractors Pump Manual for MTC rated pumps as well as all applicable requirements specified herein. Type IV pumps shall be so designed as to incorporate easy access to the impeller for quick clean out purposes. Type IV pumps shall be designed to pass spherical solids in the sizes shown in table III.

TABLE III. Solids passing sizes for type IV pumps.

Pump rating	Pump size (in inches)	Solids size (in inches)
5MTC	1.5	0.75
9MTC	2.0	1.00
15MTC	3.0	1.50
22MTC	4.0	2.00

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3 5.2 Casing. The casing, including flanged-mounted air separation chambers , if furnished, shall be cast iron or aluminum alloy in accordance with the material specified (see 6.2), except that type IV pump casings shall be furnished in aluminum alloy only. The casing shall be designed to withstand a hydrostatic test pressure equal to 1.5 times the pump discharge pressure or the total dynamic head, whichever is greater. The casing shall be equipped with a priming port and means for readily draining the casing. A standard pipe tee and plug installed on the discharge connection will be acceptable as a priming port The casing shall be designed or equipped to prevent siphonage of priming'~fiid on shutdown if a foot valve is not installed on the suction line. The pump shall be equipped with a renewable wearing surface or surfaces, or impeller adjustment so that proper clearances can be restored without the need for replacing the pump casing. The casing shall be designed with bolted plates or segmental sections to permit removal and replacement of internal pump components .

3.5.3 Impeller. The impeller shall be of the material specified (see 6.2). Imperllers shall be open or semi-open. Impellers shall be cast in one piece, balanced, and shall be keyed or screwed to the shaft in a manner to provide positive locking against any rotative loosening effect.

3.5.4 Bearings. Pumps with separate shafts coupled to the engine crankshaft shall be equipped with not less than two sets of ball or roller anti-friction bearings to support the shaft. The bearing shall be rated to withstand continuous radial or axial loading due to Inherent rotational forces and hydraulic imbalance. The bearings shall have adequate provisions for lubrication and shall be sealed against the entrance of dirt and water.

3-5-5 Shaft seal. The shaft seal shall be of the mechanical type and shall be self-lubricated, oil lubricated, or grease lubricated. The seal shall consist of a rotating element, a stationary element, a spring, and any additional elements common to particular seal designs, such as a diaphragm, bellows, spring cage, or grease cup. Stationary elements shall be fabricated of a hard, wear-resistant ceramic or brass material, or of nickel-bearing cast iron. On self-lubricated seals, the rotating wearing surface shall be carbon or a material having similar low-friction properties. Seal springs shall be stainless steel. Grease cups shall be of the spring-loaded type designed to automatically feed lubricant to the seal as required.

3.5.6 Connections. The connection sizes for the suction opening shall be of the same nominal pipe size as the discharge opening. The openings shall be tapped or fitted to provide female tapered pipe connections in accordance with ASME B 1.20.1. Both connections shall be so located or fitted that a direct-connected length of straight pipe shall extend in a horizontal direction.

3.5.6.1 Optional connections. When specified (see 6.2), the pumps shall be fitted in accordance with one of the following options:

a. Option 1: Four inch and larger pipes size pumps shall be equipped with standard class 125 pound (lb) cast iron flanges, conforming to ASME B 16.1, or class 150 lb steel flanges, conforming to ASME B 16.5, as specified (see 6.2).

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b. Option 2: Pump connections shall be fitted with straight, pipe-to-hose adapters. The hose thread shall be a male American National NH designator fire-hose coupling thread, except that 2.0 inch adapters and, when specified (see 6.2), 1.5 inch adapters shall be American National NPSH designator hose coupling threads in accordance with FED-STD-H28/10. Unless otherwise specified (see 6.2), the nominal size of the hose threads shall be the same as the pump connection.

3.5.7 Suction strainer. A suction strainer, having the same nominal pipe size as the inlet connection, shall be furnished with each pump.

3.5.8 Hose. When specified (see 6.2), hose shall be furnished with the pump unit. The hose furnished shall be four 10-foot lengths of the manufacturer's standard commercial 4-inch rubber, wire-reinforced, with hose couplings. All hoses furnished shall have the capability of being coupled together.

3.5.9 Engine. Unless otherwise specified (see 6.2), the engine shall be a commercial engine normally supplied by the contractor. Engines rated at 30 horsepower (hp) and below may be air-cooled and shall operate on unleaded gasoline or diesel fuel, as specified (see 6.2). All multi-cylinder, air-cooled engines shall be equipped with positive exhaust valve rotators and with stellite-faced exhaust valves and solid stellite valve-seat inserts. Engines rated over 30 hp shall be heavy-duty industrial diesel type, liquid-cooled, and designed to operate on fuel conforming to VV-F-800.

3.5,9.1 Horsepower. The engine, fully equipped with all accessories, shall drive the pump and operate under standard conditions at rated capacity for all applicable conditions of had and lift specified herein without exceeding the continuous hp rating of the engine. The continuous hp ratings shall be established in accordance with SAE J1349. The minimum continuous hp rating to be established shall be as specified in accordance with the CPB Contractors Pump Manual.

3.5.9.2 Engine accessories. Each engine shall be equipped with a governor, manual speed control, exhaust muffler, air cleaner, and fuel tank. All engines shall be equipped with a cooling fan with a protective screen or shroud. Diesel engines shall be equipped with a lubricating oil pressure gage, a coolant temperature gage, an oil filter, and an air cleaner (see 3.5.9.2.1). All other engine accessories supplied in commercial practice as standard equipment shall be furnished.

* 3.5.9.2.1 Diesel engine air cleaner. Air cleaner shall be of the two-stage dry-type design wherein the first stage provides centrifugal separation and the second stage provides impingement filtration with use of a suitable filter medium. When handling dust concentration of 0.025 grams per cubic foot, the air cleaner shall be capable of operating not less than 8 hours without exceeding an air restriction which could adversely affect the-rated output of the pump. When located outside the compressor housing, air intake shall be fitted with a rain cap and screen. Connections of components associated with the air cleaner installation shall be water- and gas-tight. The cleaner shall have an indicator that shows when the cartride is ready for replacement and shall be visible to the operator when the enclosure door is open.

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3 5.9 3 Starting systems. The starting system for gasoline engines rated 10 hp and below may be manual (rope, crank, or recoil) The starting system for gasoline engines rated over 10 hp shall be electric. All diesel engines shall have electric starting systems. Electric starting systems shall be 12 volts and shall include an electric cranking motor, charging alternator, regulator, dry-charged battery with cables, starter switch, and ammeter. Gasoline engines with electric starting systems shall include a high tension, distributor-type ignition system. Ether starting systems, if furnished for diesel engines, shall be of the automatic injection engine temperature limited type, or controlled from the control panel, and shall be inoperative with the engine running.

3.5.9.4 Hour meter. Engines rated over 30 hp shall be equipped with an hour meter. The meter shall either be electrical or mechanical and shall register not less than 9,999.9 hours of operating time.

3.5.9.5 Air pollution control. Pump engines shall comply with EPA 40, CFR, Part 86 governing the control of air pollution from engines intended for industrial application in effect on the date of manufacture.

3.6 Mountings.

3.6.1 Base mounting (classes 1 and 5). Base mountings shall consist of a flat steel plate or a steel plate or sheet formed to provide a raised horizontal center section with supporting vertical sections along each longitudinal edge. Base mountings consisting of a combination of steel channels, flat steel runners , or other structural steel shapes will be acceptable. The base mounting shall be of sufficient thickness and rigidity to maintain the pump and engine in proper alignment under normal conditions of operation and transport. All class 1 base-mounted pumps shall be equipped with a lifting eye or a lifting bail. Base-mounted size 5M, 5MT, and 5MTC class 1 pumps shall also be equipped with a one-man closed grip carrying handle. The maximum net weight of size 5M and 5MT shall be 85 lb. Base-mounted size 8M and 9MTC, class 1 pumps shall also be equipped with a two-man closed grip carrying handle. The maximum net weight of size 8M and 9MTC pumps shall be 170 lb. Class 5, base-mounted pumps shall be furnished with a tubular, wrap-around, welded, roll-bar frame firmly attached to the base mounting. The frame shall consist of tubular members having an outside diameter of not less than 3/4-inch. The frame shall be designed to prevent damage to any component of the pump unit should the pump roll on its top or any of its four sides. In accordance with 1.2, base mountings shall be limited to pumps having nominal capacities not greater than 9,000 gph for class 5, and not greater than 18,000 gph for class 1.

3.6.2 Skid mounting (class 3). When specified (see 6.2), skid mountings shall be furnished for pumps having nominal capacities of not less than 20,000 gph. The skid mounting shall consist of a base mounting having steel skids longitudinal along each of the two longest edges of the base mounting. The skids shall be structural steel shapes and shall be of sufficient depth to provide not less than 3-inch clearance between the ground and the underside of the base plate or cross members. The bearing surface of each skid shall be not less than 2 inches wide. Both ends of each skid shall be beveled or rounded to facilitate skidding and shall be equipped with pulling attachments. A hole, not

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less than 2 inches in diameter, near the ends of each skid shall be acceptable for pump sizes up to 20M and 3 inches in diameter for larger size units in lieu of pulling attachments. The pump and engine shall be mounted to the skid base in a manner to permit ready removal and replacement of either or both components . Skid mountings shall be designed and constructed to permit pulling (skidding) the units over dry, unpaved surfaces at speeds up to 5 miles per hour (mph) without damage to the skid, or damage to, or misalignment of any components mounted on the skid.

3.6.3 Wheel mountings. Wheel-mounted pumps, two- and four-wheel vehicular towed classes 2A, 4, and 4A shall have the tire and rim ratings conforming to the TRA Yearbook for the type and size of tires furnished. Tire and rim sizes shall be the same for all wheels. Tires for vehicular-towed pumps shall be tube or tubeless type with highway tread, and shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground. Tires for vehicular-towed pumps shall be not less than 100 level quality and shall be of domestic make. When tube type tires are furnished, tubes shall be heavy-duty type, and shall be of proper size for the tires furnished.

3 6 3.1 Manually-propelled (class 2). Class 2, wheel-mounted, manually-propelled pumps shall have wheel mounting requirements as specified in table IV. Class 2 mountings shall consist of a two-wheel, rubber-tired cart with a push bar or handle, and a supporting leg. Tires shall be of the semi-pneumatic industrial type, or pneumatic wheelbarrow type in accordance with table IV (see 6.2). Tires and wheels shall be commercially rated to withstand speeds up to 10 mph when supporting the gross weight of the pump unit.

3.6.3.2 Vehicular-towed (classes 2A and 4) wheel-mounted. Wheel mountings for class 2A and class 4 pumps shall be designed for vehicular towing over graded gravel roads at speeds up to 20 mph with wheel and tire sizes specified in 3.6.3. The undercarriage shall include a drawbar and lunette-type towing ring. The towing ring shall be fabricated of l-inch round steel stock, and shall form a ring with 3-inch ±1/16-inch inside diameter The towing assembly shall include two safety chains with links formed of stock not less than required by DoT FMVSSR. The safety chains shall be anchored to the drawbar and shall extend 24 inches beyond the towing ring. The loose end of each chain shall be fitted with a safety-type hook or a pin-type clevis. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with wheeled-undercarriages loaded with rated payload. Class 2A, two-wheeled under carriages shall be equipped with one or store retractable landing legs located, as required, to prevent the unit from tipping when disconnected from the towing vehicle. The leg(s) shall be capable of maintaining the unit in a stable, level attitude during operation. Class 4 wheeled undercarriages, applicable to sizes 90M and 70MT, and larger rated pumps on an optional basis shall be furnished with four rubber-tired wheels and automotive type steering. Clearances shall preclude interference between tires and any other part of the assembly when making turns and under the operating conditions specified herein.

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TABLE IV. Wheel mounting requirements for class 2. manually propelled.

Pump rating	Minimum tire sizes	
	Semi-pneumatic Industrial	Pneumatic Wheelbarrow
5M, 5MT, 5MTC	10 X 2.75	
8M, 9MTC	10 X 2.75	
10M, 10MT	10 X 2.75	
15M, 18M, 18MT, 15MTC	10 X 2.75	4.00 X 8 1/
20M, 22MTC		4.00 X 8
40M		4.00 X 8 2/

1/ Optional: To be furnished when specified (see 6.2).
2/ To be furnished when class 2 is specified in lieu of class 2A for size 40M (see 6.2).

3.6.3.3 Vehicular-towed (class 4A) wheel-mounted. The four-wheeled tandem axle undercarriage for class 4A pump units shall be applicable to sizes 90M and 70MT and larger rated pumps on an optional basis. Wheel mountings shall be designed for vehicular towing up to speeds of 50 mph on improved roads and over graded gravel roads at speeds up to 20 mph with wheel and tire sizes -specified in 3.6.3. The undercarriage shall include a drawbar and lunette-type towing ring adjustable from 20 to 33 inches above the ground. The towing ring shall be fabricated of l-inch round steel stock and shall form a ring with 3-inch ±1/16-inch inside diameter. The towing assembly shall include two safety chains with links formed of stock not less than required by DoT FMVSSR. The safety chains shall be anchored to the drawbar and shall extend 24 inches beyond the towing rings. The loose end of each chain shall be fitted with a safety hook or a pin type clevis. The tandem axle ratings shall be not less than equal to the load imposed on each axle, measured at the ground level when loaded with rated payload. The tandem axle undercarriage shall be furnished with the manufacturer's standard suspension system. Each component of the suspension system shall have a rated capacity not less than that equal to the load imposed, measured at the ground when the trailer is loaded with its rated payload. The wheels and axles shall be equipped with brakes conforming to DoT FMVSSR. The undercarriage shall be equipped with one or more retractable landing legs located, as required, to prevent the unit from tipping when disconnected from towing vehicles. The leg(s) shall be adjustable to maintain the unit in a stable, level attitude during operation. Clearances shall preclude interference between tires and any other part of the undercarriage under the operating conditions specified herein.

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3.7 Lifting and tiedown attachments. Class 1, size 5M, 5MT, 5MTC, 8M, and 9MTC pumps shall be equipped with lifting attachments as specified in 3.6 1 All other pumps shall be equipped with lifting and tiedown attachments conforming to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the pump unit. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tie down of the pump unit on the carrier when shipped.

3.8 Safety. All mechanical and electrical parts that are -of such nature or so located as to become a hazard to operating or maintenance personnel shall be enclosed or properly guarded. The pump shall comply with OSHA 29, CFR, 1910.95 for noise and 1910.219 for guarding.

3.8.1 Noise for Air Force acquisition. When specified (see 6.2), for Air Force acquisition, the noise level at the operator's position at the control panel, while the pump is delivering rated capacity, shall be within the limits for the preferred frequencies listed in MIL-STD-1474, category D. If the noise level is not within acceptable limits, the distance from the control panel at which it becomes acceptable shall be determined. That distance shall be shown on warning plates mounted on both sides of each pump unit, stating: "WARNING , EAR PROTECTION REQUIRED WITHIN ___ FEET WHEN EQUIPMENT IS OPERATING" (see 3.12). The manufacturer of the pump shall be responsible for filling in the number of feet for the required distance. The warning shall be legible from the distance shown on the warning plate.

3.8.2 DoT FMVSSR. All vehicular-towed pump units shall comply with all applicable DoT FMVSSR in effect on date of manufacture.

3.9 Performance requirements .

3.9.1 Capacity. The pump, with the horizontal centerline of the impeller 10, 15, 20, and 25 feet above the water level, shall meet the minimum capacity requirements specified in the CPB Contractors Pump Manual for Pi, MT, and M, rated pumps, as applicable. The capacity requirements specified for each type pump shall be attainable without exceeding the continuous hp of the engine as specified herein. Pump capacities at the specified heads and static suction lifts shall be with the contractor's standard suction hose 5 feet longer than the static suction lift. The specific weight of the water at a temperature of 68 degrees Fahrenheit (^{o}F) at sea level (barometric pressure of 29.92 inches of mercury) shall be 62.3 pounds per cubic foot.

3.10.2 Priming. After initial filling with priming fluid, the pump shall, on succeeding startups, automatically self-prime, as static suction lifts up to 25 feet under the standard conditions of fresh water temperature and atmospheric pressure specified in 3.9.1 for capacity determinations.

3.9.3 Solids handling. The pumps shall pass solids as specified in 3.5.1.1, 3.5.1.2, and 3.5.1.3, as applicable for the pump rating and size required, without clogging the impeller or pump passages

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3.9 4 Mobility.

3.9.4 1 Skid-mounted pumps. All skid-mounted pumps shall be capable of being towed at 5 mph in a series of S-curves for not less than 300 feet. There shall be no damage to the skid or damage to or misalignment of the pump mounted on the skid,

3.9.4.2 Vehicular-towed pumps. All vehicular-towed pumps mounted on automotive-type tires shall be capable of being towed at 20 mph over graded gravel roads for a distance of not less than 15 miles. There shall be no damage to or misalignment of the pump.

3.9.5 Lifting devices. The lifting devices, except for one-man or two-man close grip carrying handles, shall be capable of withstanding a loading equal to 2.5 times the normal static load when the pump is suspended in a normal position.

3.9.6 Hydrostatic pressure. The casing of the pump shall be capable of withstanding a hydrostatic pressure equal to 1.5 times the pump discharge pressure or the total dynamic head, whichever is greater. There shall be no damage or permanent deformation of the pump casing or leakage in the casing or through the casing gasket.

3.9.7 Engine horsepower. The engine shall operate as specified in 3.5.8.1.

3.9.8 Environmental The pump shall be capable of starting and operating not necessarily at rated capacity, in ambient temperatures up to 120°F at sea level and 11O°F at a 5,000 foot (barometric pressure of 24.9 ±0.2 inches of mercury) elevation. The pump shall also be able to withstand extended periods of storage at temperatures to - 25°F without adverse affect on the subsequent operability or efficiency.

3.10 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.11 Identification plate. An identification plate will be furnished by the contracting officer for each pump unit. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each pump unit in a conspicuous place with nonferrous screws, rivets, or bolts not less than l/8-inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.12 Instruction plates. The pump unit shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than l/8-inch diameter.

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3.13 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be in accordance with FED-STD-595 and the color number shall be as specified (see 6.2) . Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease , welding slag and spatter, loose will scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to ensure the adhesion of the coating system, The painting shall consist of at least one coat of primer and one finish coat of acrylic-based enamel. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 roils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.14 Rustproofing. The wheeled undercarriage of vehicular-towed pumps shall be rust-proofed in accordance with FED-STD-297.

3.15 Lubrication Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high-pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The unit shall be lubricated as specified in section 5.

3.16 Workmanship. The quality of workmanship shall be such as to produce pump units that are in accordance with the requirements of the CM Contractors Pump Manual and as specified herein to insure proper functioning of all parts of the pump.

3,16.1 Steel fabrication. The steel used in fabrication shall be free from kinks , sharp bends, and other conditions which would be dileterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape.

3.16.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.16.3 Riveted connection. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes." Rivet heads, when not countersunk Or flattened, shall be of approved shape and uniform size for the same diameter Of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

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3.16.4 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

3.16.5 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter.. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one pump unit when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4, the tests of 4.5, and when specified (see 6.2), the preproduction pack inspection of 4.6. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract. In the event the contractor desires to deliver the test model as

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a contract item, it shall be delivered as the last item on the contract only after the contractor, at his own cost and expense, shall have completely cleaned, reconditioned, and overhauled the unit, making such replacements and modifications thereto as are required to make the unit acceptable as a contract item.

4.2 2 Quality conformance inspection The quality conformance inspection shall include the examination of 4.4, the tests of 4.5.1, 4.5 2, 4,5.3, 4.5.4, 4.5.7, and 4.5.8, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3, except that all pump units produced shall be tested for the operational test of 4.5.1 and the hydrostatic pressure test of 4.5.7.

* 4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one pump unit. All pump units offered for delivery at one time shall be considered a lot for the purpose of inspection. Guidance for inspection level and Acceptable Quality Level (AQL) is provided in 6.6,

4.4 Examination. Each pump unit shall be examined for compliance with the requirements specified in section 3 of this document. Any redesign or modification of the contractor's standard product to comply with specified requirements , or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Tests. Prior to the tests specified herein which require the pump units be operated, the pump units shall be furnished with lubricants recommended by the manufacturer for use in the ambient temperature at the test location. Batteries used for test purposes on pump units equipped with electric starting systems shall be supplied by the manufacturer for use in lieu of the dry-charged battery that is co be shipped with the pump unit.

4.5.1 Operational test . The pump shall be run-in for a period of not less than one hour to satisfy the operability of the pump. Failure of any of the following listed items to meet the manufacturer's specifications shall constitute failure of the operational test:

a. Alignment of pump and engine.
b. Operation of mechanical seals and lubrication systems.
c. Freedom of liquid passages.
d. Operability of engine governor and manual speed control.
e. Proper functioning and calibration of engine meters and gages.
f. Any condition evidenced by excessive temperature, vibration, or noise.

4.5.2 Capacity-head test. The pump shall be tested to verify compliance with 3.9.1. The tests shall be performed in accordance with CPB Contractors Pump Manual. Failure of the pump to meet any capacity-head requirements shall constitute failure of the test.

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4.5.3 Priming test. The pump shall be tested to verify compliance with 3.9 2 The pump shall be operated at a speed not greater than the engine manufacturer's recommended continuous operating hp or at the minimum operating hp specified in 3.9.7, whichever is less.

4.5 4 Solids handling test. The pump shall be tested to verify compliance with 3.9.3 for the pump rating and size required. Failure of the pump to meet the solids handling requirements shall constitute failure of the test.

4.5 5 Mobility.

4.5.5.1 Skid-mounted pumps. Skid-mounted pumps shall be tested to verify compliance with 3.9.4.1. The skid-mounted pump shall be towed at 5 mph in a series of S-curves for not less than 300 feet. hy evidence of damage to the skid or damage to or misalignment of the pump mounted on the skid shall constitute failure of the test.

'4.5.5.2 Vehicular-towed pumps. Vehicular-towed pumps shall be tested to verify compliance with 3.9.4.2. Pumps that are mounted on automotive-type tires shall be towed at speeds up to 20 mph over graded gravel roads for a distance of not less than 15 miles. by evidence of structural deformation, failure of the mounting, damage or misalignment affecting serviceability of either the engine or pump shall constitute failure of the test.

4.5.6 Lifting devices test. The pump unit shall be tested to verify conformance with 3.9.5. A static load equal to 2.5 times the normal static load of the pump unit shall be applied to the lifting device, except for one-man or two-man carrying handles. Weld failure or permanent deformation of the lifting device or structural member shall constitute failure of the test.

4.5.7 Hydrostatic pressure test. All pump casings produced shall be tested to verify compliance with 3.9.6, All pump casings shall be subjected to a hydrostatic pressure equal to 1.5 times the pump discharge pressure or the total dynamic head, whichever is greater. Minor leakage through the mechanical seal shall not be cause for rejection. Any damage or permanent deformation to the pump casing or leakage in the casing or through the casing gasket shall constitute failure of the test.

4 5.8 Engine horsepower test. For commercial engines, the pump manufacturer shall make available, when requested, performance curves showing hp ratings and operating speeds for the engine being furnished. The curves shall be based on dynometer tests certified by the engine manufacturer.

4.5.9 Environmental tests. The pump shall be tested to verify compliance with 3.9.8. The pump shall be tested in accordance with the applicable test methods of MIL-STD-1400 to meet the environmental conditions specified in 3.9.8. Failure of the pump to operate as specified herein during and after exposure to the environmental conditions shall constitute failure of the test.

4.6 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-P-106O3. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

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5 PACKAGING

5.1 Preservation-packaging. Preservation, packing, and marking shall be in accordance with the requirements of MIL-P-106O3 with the level of preservation and the level of packing as specified (see 6.2).

5.2 Landing gear. . Wheel-mounted pump units shall not be shipped resting on landing gear.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The type I, 111, and IV pump units covered by this specification are intended for dewatering operations on construction jobs and for other applications in general construction work where a portable self-priming pump is required. Class 5 pumps, with wrap-around tubular frames, can be used to advantage as emergency fire fighting pumps when equipped with the proper hose adapters. Cast-iron impellers are best suited for the pumps in dewatering operations. The higher surface hardness of cast-iron impellers enables these impellers to withstand the abrasive action of sedimentary solids normally encountered. The HI recommends cast-iron pumps with bronze impellers (material B) when pumping salt water. When specific capacities are required at known head, tables in the CPB Contractors Pump Manual should be consulted. See section 2 for address of CPB where copies of the manual may be obtained.

* 6.1.1 Selection factors.

a. Fungus resistant varnish conforming to MIL-V-173 should be used to coat electrical components and circuit elements, including terminal and circuit connections, when the unit is to be installed in humid conditions. Components and elements inherently inert to fungi or in hermetically sealed enclosures or current-carrying contact surfaces should not be coated
b. Electromagnetic interference suppression (EMI), when required, should conform to the EMI suppression requirement-s and test limits as specified in SAE J551.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title , number, and date of this specification.
b. Type, size, material, class, and style of pump unit required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)"
d. When first article is required for inspection and approval (see 3.2 2 and 4.2.1).
e. Material required for pump casing (see 3.5.2). 2).
f. Material required for impeller (see 3.5.3).

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g. When optional connections are required; type of flange for option 1; when 1.50 inch thread adapters are to be NPSH in lieu of NH for option 2; when size of hose threads are other than as specified for option 2 (see 3.5.6.1).
h. When hose is to be furnished (see 3.5,8),
i. Type of engine, if other than as specified, and whether engine is to be gasoline or diesel driven (see 3 5 9)
j When skid mountings are required for class 3 pumps with capacities over 20,000 gph (see 3.6.2).
k. When pneumatic wheelbarrow type tires are required for sizes 15M, 18M, 18MT, 15MTC, and 40M (see 3.6.3.1).
1. When noise level limits of MIL-STD-1474 are to be invoked for Air Force acquisition (see 3.8.1).
m. Color of finish coat required (see 3.13).
n. When means of lubrication is other than as specified (see 3.15).
o. When a preproduction pack inspection is required and the time frame required for submission (see 4.2.1 and 4.6).
p. Level of preservation and level of packing are required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations (FAR) Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4 2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Standard lubricants, When requested by the manufacturer, the contracting officer should obtain through established channels, a list of standard military lubricants applicable to the equipment covered herein. This information is contained in the Department of Defense Section of the Federal Supply Catalog, 9100-IL.

* 6.6 Sampling procedures .

* 6.6.1 Sampling for examination. Recommended inspection level is II and AQL is 2.5 (see 4 3).

* 6.6.2 Sampling for tests. Recommended inspection level is S-4 and AQL is 2.5 (see 4,3).

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6.7 Part or identifying number (PIN). The PIN to be used for pump units applied to this specification are created as follows:

	M17861	XXX	X	XX
Specification part number				
Type and size code				
Material code				
Style and class code				

6.7.1 Type and size to code identifier. The type and size of the pump unit is identified by a three-digit identifier as shown in table V. The first number identifies the type; the second and third numbers identify the size.

TABLE V. Type and size to code identifier.

Type and size	Code number	Type and size	Code number	Type and size	Code number
Type I		Type III		Type IV	
Size 5M	101	Size 5MT	331	Size 5MTC	441
Size 8M	102	Size 10MT	332	Size 9MTC	442
Size 10M	103	Size 18MT	333	Size 15MTC	443
Size 15M	104	Size 33MT	334	Size 22MTC	444
Size 18M	105	Size 35MT	335		
Size 20M	106	Size 70MT	336		
Size 40M	107				
Size 90M	108				
Size 125M	109				
Size 200M	110				

6.7.2 Material to code identifier. The material of the pump unit is identified by the single letter specified (see 1.2).

6.7.3 Style and class to code identifier. The style and class of the pump unit is identified by a two-digit identifier as shown in table VII. The first character is a single letter identifying the style and the second character is a single number identifying the class.

TABLE VII. Style and class to code identifier.

	Class						
Style	1	2	2A	3	4	4A	5
A	A1	A2	A3	A4	A5	A6	A7
B	B1	B2	B3	B4	B5	B6	B7

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6 8 Cross-reference of classification. This revision contains no changes in the classification of the pump units. The following data is a historical record of classification changes made in previous revisions of this specification:

MIL-P-17861C	MIL-P-17861D
Type I	Type I
Size 5M	Size 5M
Size 8M	Size 8M
Size 10H	Size 10M
Not designated	Size 15M
Size 17M	Deleted
Not designated	Size 18M
Size 20M	Size 20M
Size 30M	Deleted
Size 40M	Size 40M
Size 90M	Size 90M
Not designated	Size 125M
Not designated	Size ZOOM
Type II	Deleted
Size 4H	Deleted
Size 10H	Deleted
Size 15H	Deleted
Size 40H	Deleted
Size 80H	Deleted
Not designated	Type III
Not designated	Size 5MT
Not designated	Size 10MT
Not designated	Size 18MT
Not designated	Size 33MT
Not designated	Size 35MT
Not designated	Size 70MT
Not designated	Type IV
Not designated	Size 5MTC
Not designated	Size 9MTC
Not designated	Size 15MTC
Not designated	Size 22MTC
Material A	Material A
Material B	Material B
Material C	Material C
Material D	Material D
Class 1	Class 1
Class 2	Class 2
Not designated	Class 2A
Class 3	Class 3
Class 4	Class 4
Not designated	Class 4A
Class 5	Class 5

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* 6.9 Subject term (key word) listing.

Pump
Solids handling
Skid mounted
Base mounted
Wheel mounted

6.10 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections , deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Navy - YD
Air Force - 99

Review Activity:
DLA - CS

User Activities:
Army - ME, CE
Navy - MC

Preparing Activity:
Navy - YD

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-P-17861F	2. DOCUMENT DATE (YYMMDD) 910930

3. DOCUMENT TITLE
PUMP UNIT, CENTRIFUGAL: PORTABLE, SELF-PRIMING, ENGINE-DRIVEN (DEWATERING; CONTRACTOR AND

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME NNE L. LYONS	b. TELEPHONE *(Include Area Code)* (1) Commercial: (805) 982-5761 (2) AUTOVON: 551-5761
c. ADDRESS *(Include Zip Code)* Commanding Officer (156) Naval Construction Battalion Center Port Hueneme, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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